Privacy by Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Privacy by Design before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Red Team the Product

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Quality Rubrics

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Requests Responses and JSON

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Privacy by Design trace and record the first step you can explain without help.

Use W2: Trace the worked case for Data **retention** and deletion. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Privacy by Design explanation against the supplied facts before discussing it.

Use the worked case for Data **retention** and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Retention is5 days, inclusive **deletion boundary**. Ages4, 5, 6; a log stores complete content. Trace required action.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Privacy by Design target to a trace and a justified conclusion.

A study tool needs topic and reading level but proposes storing addresses and all conversations forever. Design a smaller record and a deletion rule.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Privacy by Design case and show the deciding evidence.

A study tool plans to retain names, addresses and all chats forever.

Design a smaller data plan.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A practice quiz needs a topic and difficulty. A form also asks for home address and date of birth. What should be removed and why?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 28 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Privacy by Design adds to Verified Study Buddy and keep the evidence.

Create a minimal data map and visible memory controls before expanding features.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A study tool needs topic and reading level but proposes storing addresses and all conversations forever. Design a smaller record and a deletion rule.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Privacy by Design facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Privacy by Design", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Privacy by Design facts and keep your reasoning.

Record ages are6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Privacy by Design facts and keep your reasoning.

A report omits this qualification: This fictional **retention** exercise is not a statement of legal **retention** requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Privacy by Design facts and keep your reasoning.

Retention is5 days, inclusive **deletion boundary**. Ages4, 5, 6; a log stores complete content. Trace required action.

My reasoning and evidence

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Privacy by Design facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Privacy by Design", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Privacy by Design

> Trace deletion through both a working folder and a cache. Explain the labelled step before checking the changed condition.

Trace deletion through both a working folder and a cache.

Fictional **retention** policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/28

Privacy by Design

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare ages

A is below 3; B is exactly 3; C is above 3.



2. Delete eligible

Keep A. Delete B and C from the working folder.



3. Check copies

Delete cached B too; otherwise the declared content remains retained.

CHANGE ONE THING

The cache deletion fails.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Trace deletion through both a working folder and a cache.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Delete cached B too; otherwise the declared content remains retained.

1. Compare ages

2. Delete eligible

3. Check copies

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check copies Delete cached B too; otherwise the declared content remains retained. Read the picture in order.

Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The cache deletion fails. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Report incomplete deletion rather than success.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report incomplete deletion rather than success. Removing the working copy does not remove the known retained cache copy.

Repair a missing building block

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Explain the **mechanism** and terminology of system evaluation sets. Short practice: Why is “one out of two checks passed” an insufficient release decision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

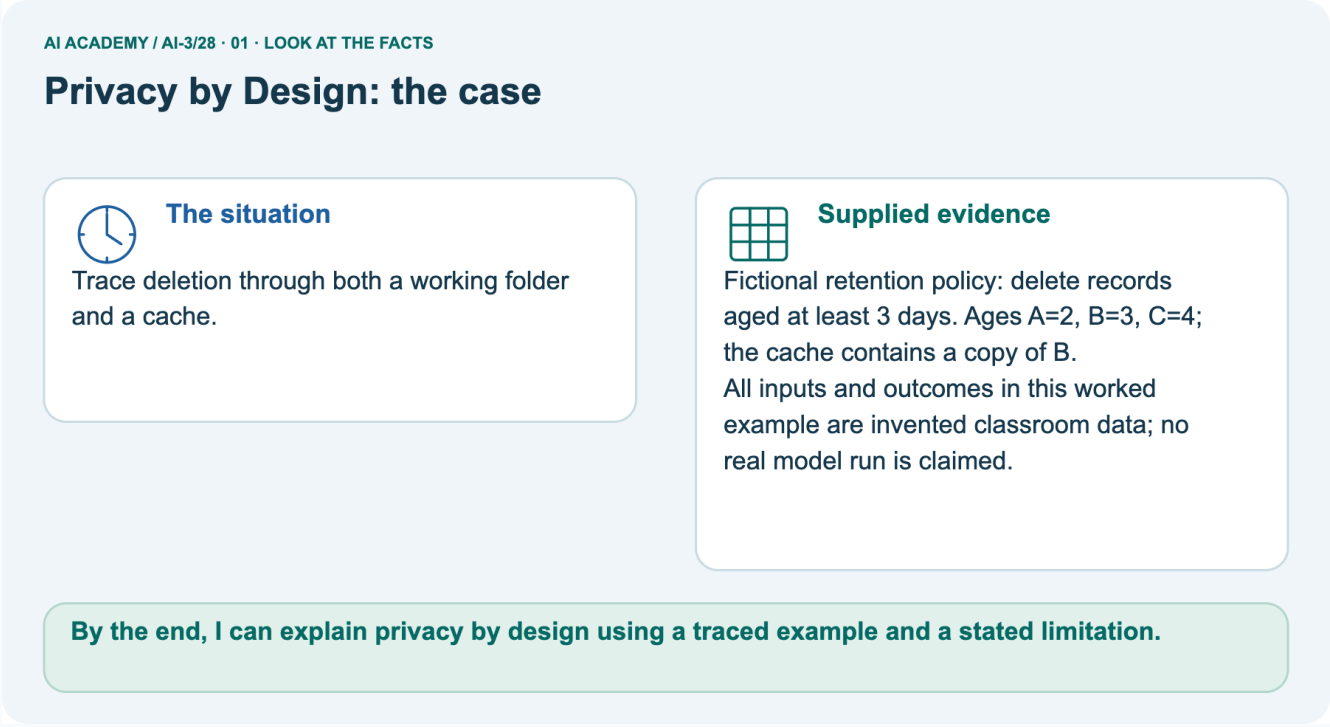
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Privacy by Design picture from its labels, then record a relevant limit.

01. Privacy by Design: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

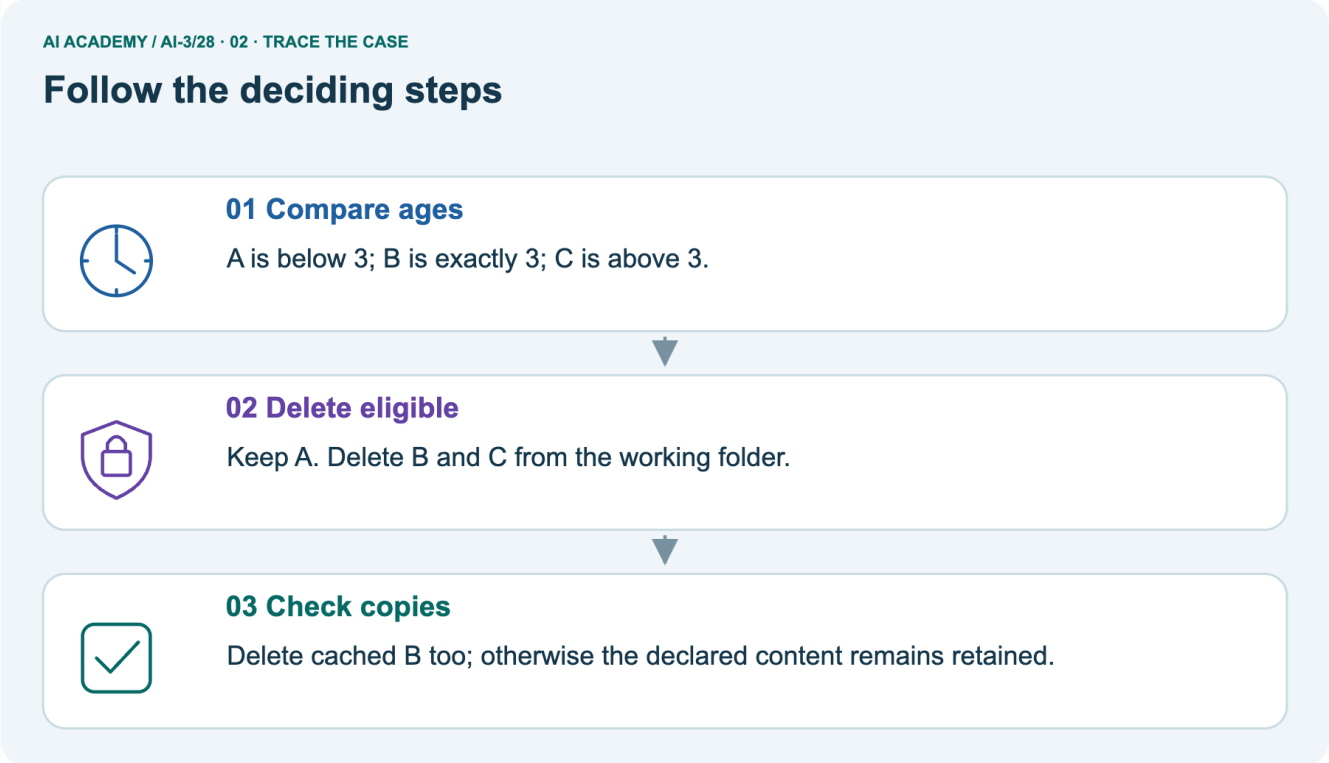
Trace deletion through both a working folder and a cache. Supplied facts: Fictional **retention** policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



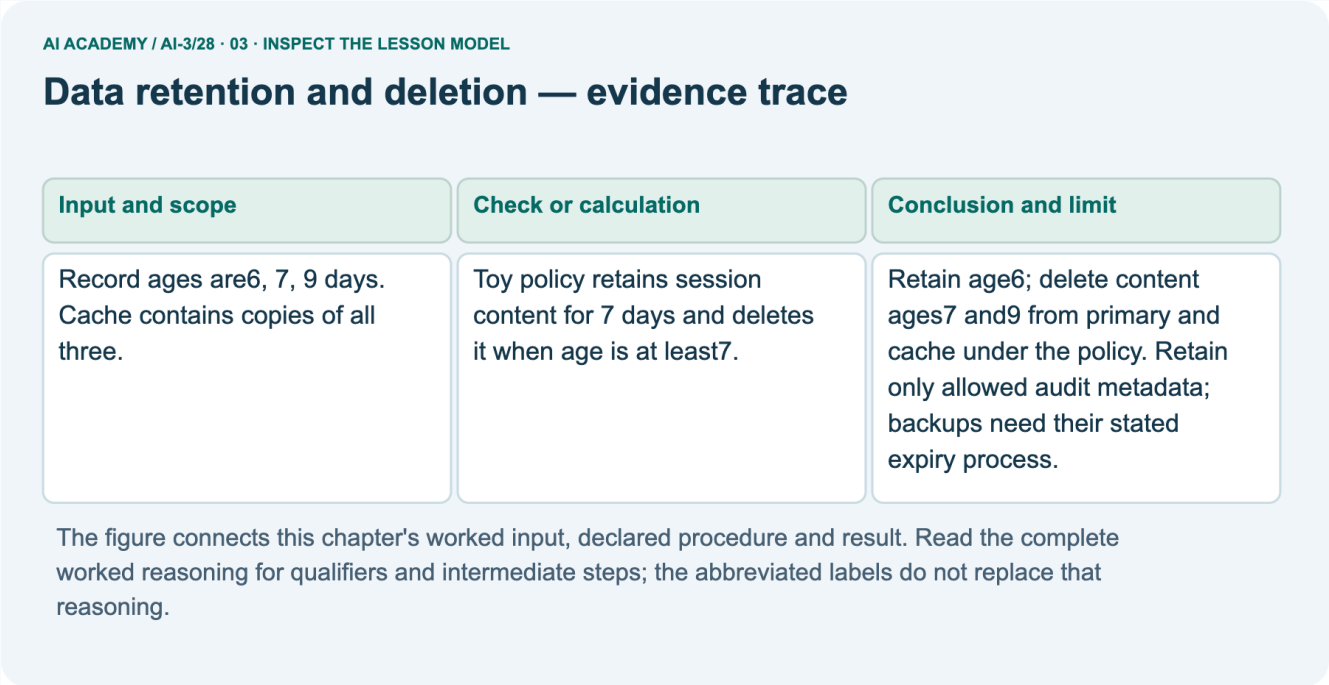
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare ages: A is below 3; B is exactly 3; C is above 3. → Delete eligible: Keep A. Delete B and C from the working folder. → Check copies: Delete cached B too; otherwise the declared content remains retained.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Data retention and deletion — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Data **retention** and deletion — **evidence** trace. Record ages are6, 7, 9 days.

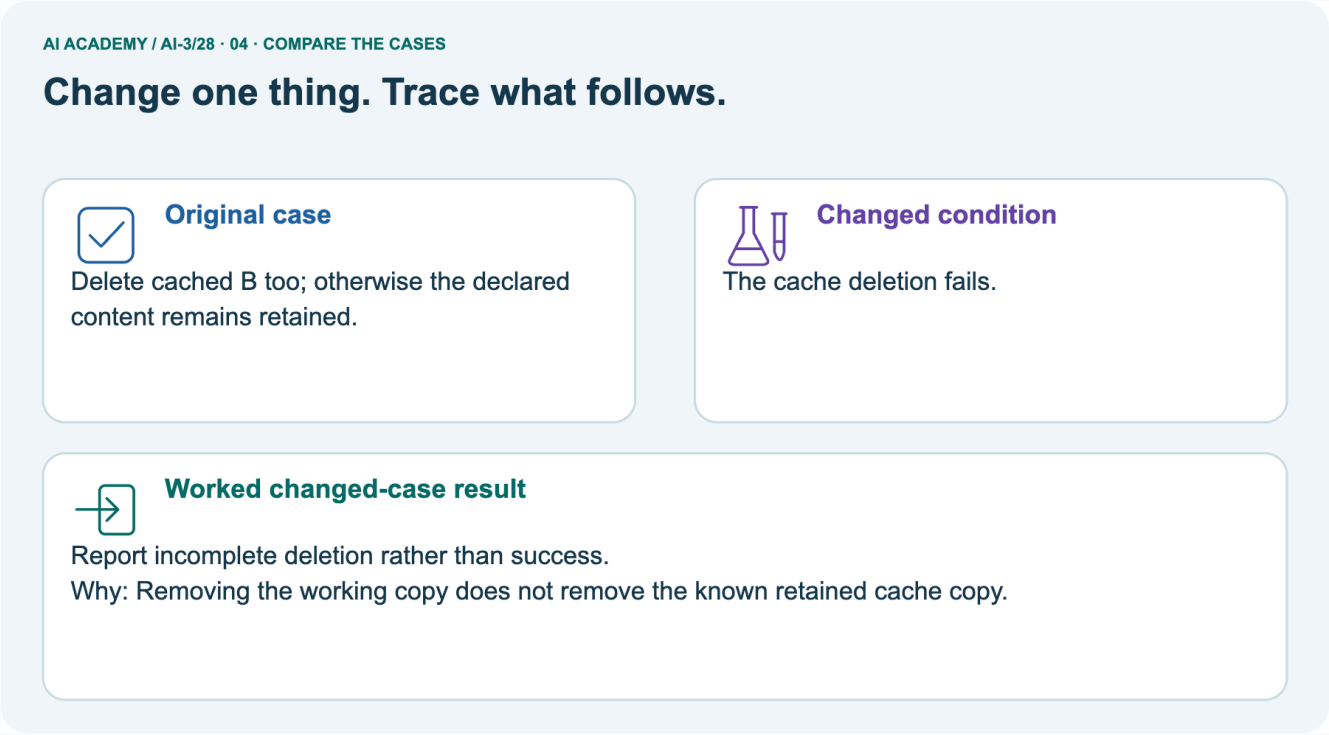
Cache contains copies of all three. | Toy policy retains session content for 7 days and deletes it when age is at least7. | Retain age6; delete content ages7 and9 from primary and cache under the policy.

Retain only allowed audit metadata; backups need their stated expiry **process**.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

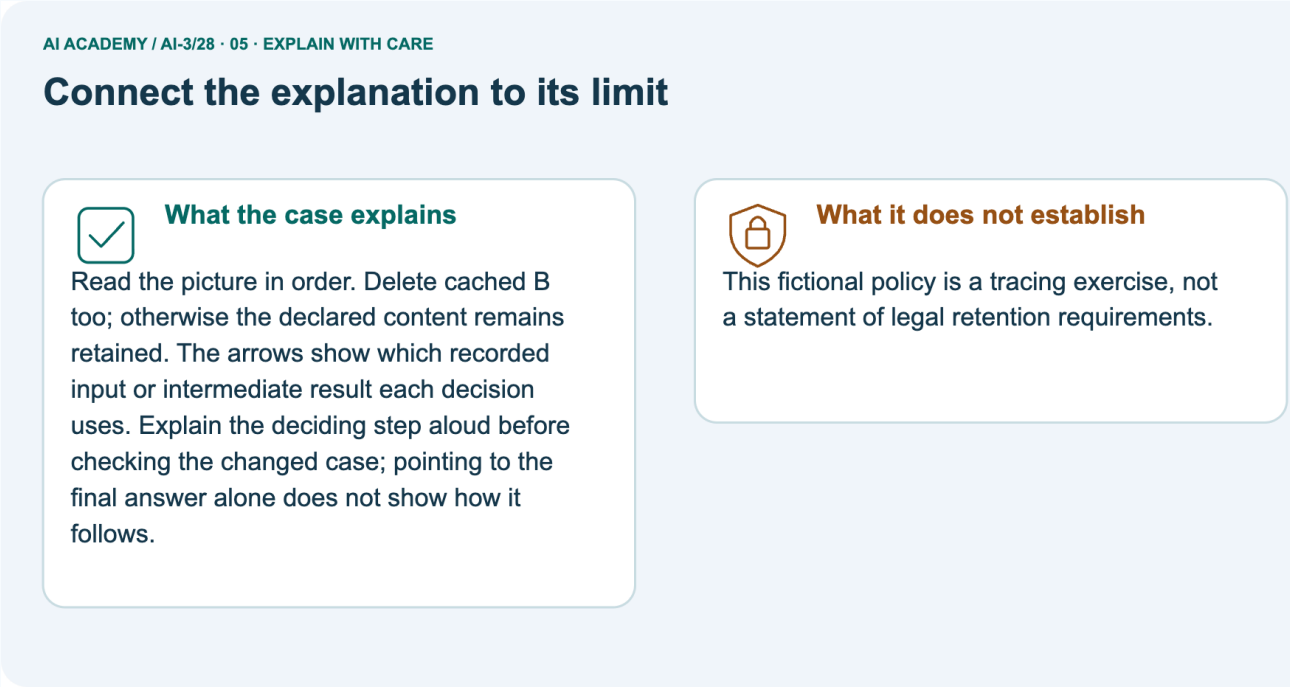
Original case: Delete cached B too; otherwise the declared content remains retained. Changed condition: The cache deletion fails. Expected result: Report incomplete deletion rather than success.

Reason: Removing the working copy does not remove the known retained cache copy.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

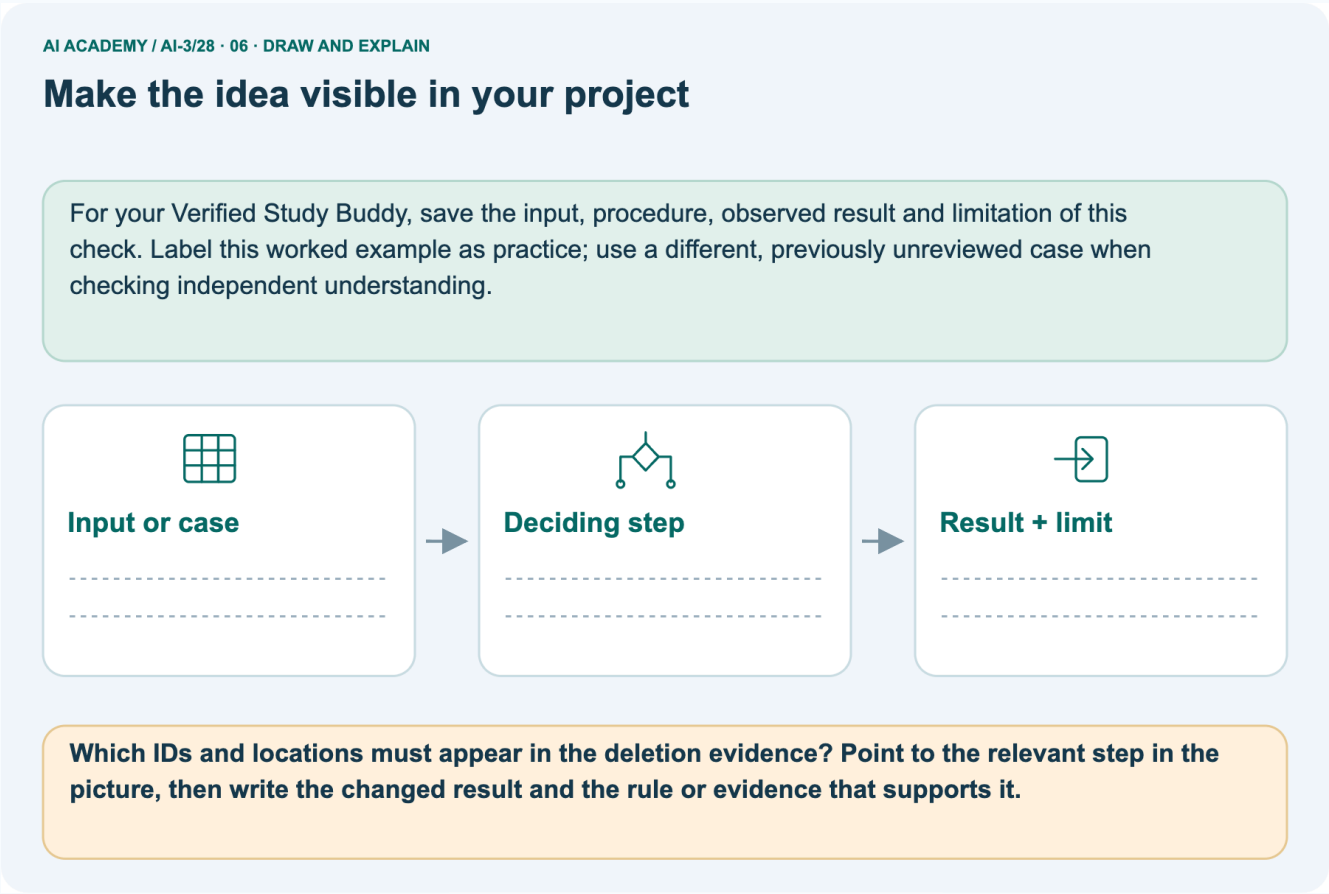
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This fictional policy is a tracing exercise, not a statement of legal **retention** requirements.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.